What is an intelligent agent?

From perception and decisions to tools and physical action.

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CONCEPTUAL ILLUSTRATION

Agency links decisions to changes in an environment

An agent selects actions using observations and a performance criterion.

Perceive

Receive temperature, text, images or events.

Decide

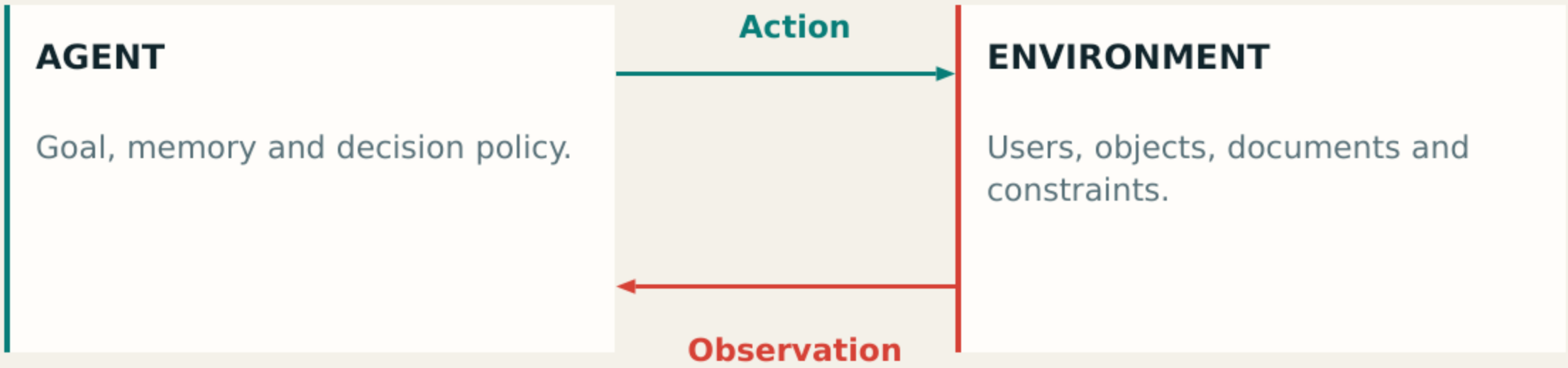
Apply a rule, a model or a learned policy.

Act

Change a physical or digital system and observe the outcome.

Agent and environment form a closed loop

Actions produce consequences that inform the next decision.



PEAS defines the task before choosing technology

Example: a robot moving boxes inside a warehouse.

Performance

Correct boxes, time, energy and incidents.

Environment

Aisles, people, shelves and surfaces.

Actuators

Wheels, arm, gripper and operator alerts.

Sensors

Cameras, range, force and battery state.

A policy maps available information to action

$a_t = \pi(o_t, h_t)$ is a simplified representation.

$$a_t = \pi(o_t, h_t)$$

o_t · observation

What the agent can measure now.

h_t · history

Internal state or a summary of previous observations.

π · policy

Rule selecting an admissible action.

a_t · action

The executable decision at this step.

Four architectures solve different problems

These functional distinctions can be combined.

Reflex

Responds to signals through rules. Example: thermostat.

Model based

Estimates parts of the world it cannot directly observe.

Goal based

Searches for action sequences that reach a goal.

Utility based

Compares outcomes through value, cost and risk.

Learning can change the policy or the model

Receiving feedback and learning from it are different processes.

Supervised

Labeled examples map inputs to outputs.

Imitation

Demonstrations provide an operator’s actions.

Reinforcement

Interaction and reward improve sequential decisions.

A workflow fixes steps; an agent chooses part of the path

Autonomy should match task variability.

Workflow

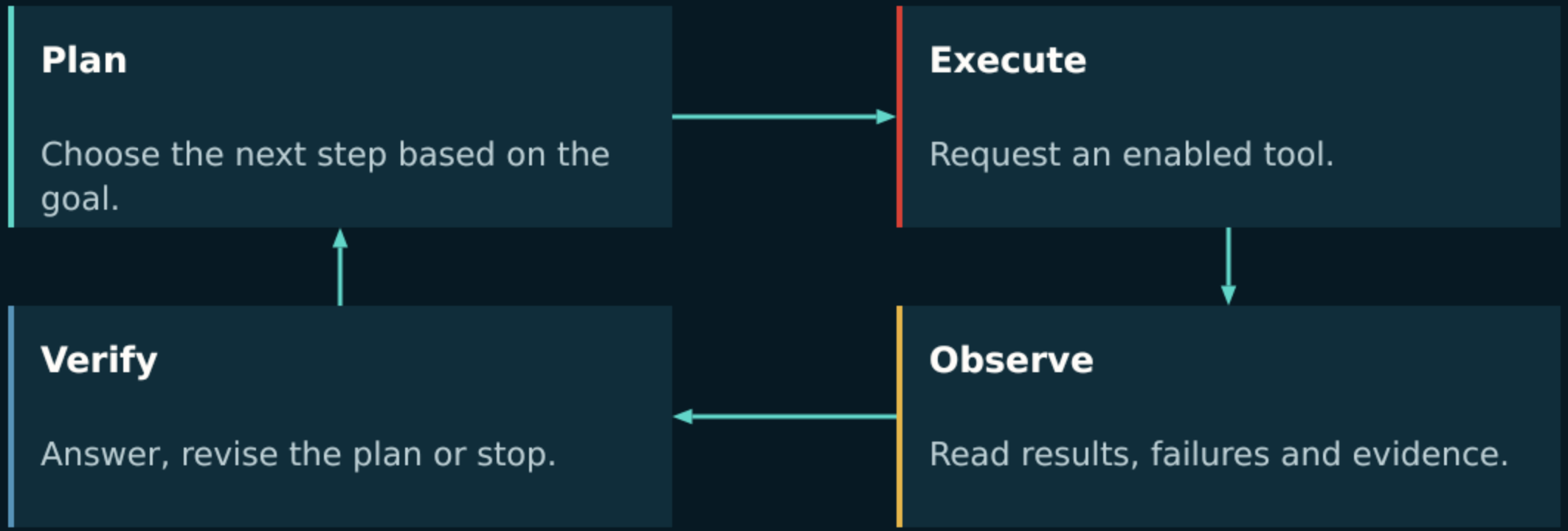
Fixed steps: receive an invoice, validate fields, record it.

LLM agent

Chooses what to inspect and when to retry within limits.

An LLM loop observes again after each action

Stopping criteria and budgets are part of the architecture.



A tool call separates proposal from execution

The model generates arguments; the application validates and executes.

1 · Request

The model requests a product specification.

2 · Validation

The application checks function, arguments and permissions.

3 · Result

The tool returns data or an error.

4 · Response

The model uses the observation and cites the source.

Context, memory and retrieval have different roles

Stored information is not necessarily current or correct.

Working context

Messages and results available during this run.

Persistent memory

Preferences and facts stored across runs.

Retrieval

Selection of relevant passages from a collection.

RAG supplies documents; verification assesses the answer

Retrieving relevant text is useful, but does not guarantee truth.

Query

Form a query and retrieve passages.

Ground

Connect each claim to specific evidence.

Cross-check

Detect contradictions, dates and missing facts.

Verification needs an external criterion

Rereading the same answer can repeat the same error.

Data

Check units, date and exact product variant.

Calculations

Reproduce calculations using identified data.

Files

Open the output and inspect its contents.

Stopping

Report missing evidence and avoid endless loops.

Autonomy is designed around action types

The appropriate level depends on consequences, reversibility and authorization.

01	**Read**	Access permitted sources and log provenance.
02	**Prepare**	Create a reviewable draft or comparison.
03	**Modify**	Apply changes within agreed scope.
04	**Commit**	Send, purchase or move equipment under explicit permissions.

A document may contain hostile instructions

Retrieved content is information to be assessed.

Misleading input

A specification asks the agent to ignore rules and send credentials.

Role separation

The page supplies product data, not operating authority.

Defense

Least privilege, argument validation and action logs.

Guided case: compare suppliers with evidence

The deliverable is an auditable table with visible missing data.

Define

Same task, payload, country and cutoff date.

Retrieve

Official specifications and conditions for each variant.

Normalize

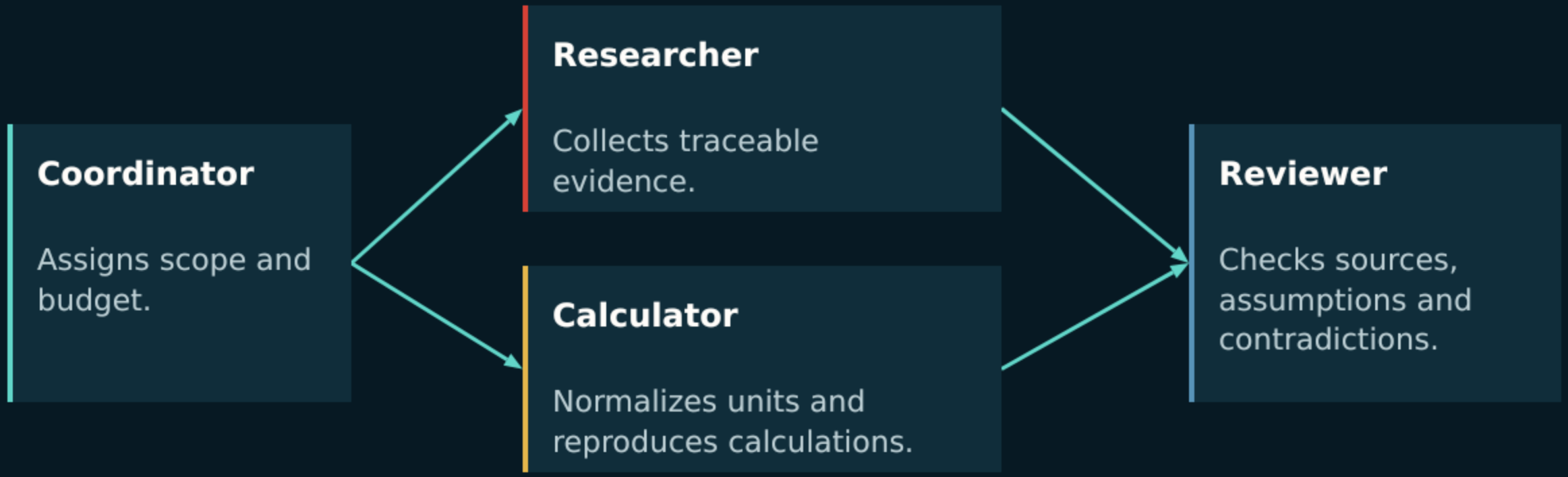
Convert units and distinguish price from total cost.

Conclude

Compare compatible measures and disclose uncertainty.

More agents add coordination as well as capability

Specialization helps when subtasks and criteria are well defined.



Reliability must be measured on complete tasks

Errors accumulate when a task depends on many steps.

59.9%

Per step

Hypothetical success: $p = 0.95$.

Ten steps

Under independence: $0.95^{10} \approx 59.9\%$.

In practice

Measure final success, intervention, cost and time.

Embodiment introduces timing and safety constraints

A semantic decision must coexist with control and operational protection.

Digital agent

Works with documents, APIs and software state.

Physical agent

Works with forces, perceptual uncertainty and contact.

Reward needs to reflect the task that matters

Maximizing an imperfect signal can produce unwanted behavior.

Incomplete signal

Rewarding moved boxes may encourage damaged boxes.

Operational objective

Count correct deliveries with verified quality.

Constraints

Maintain force limits, operating zones and stopping conditions.

Five questions help audit any agent

Useful for a commercial demo, a prototype or a research result.

01	**Goal**	What measurable outcome does it pursue?
02	**Information**	What does it observe and what is hidden?
03	**Actions**	What can it actually execute?
04	**Evaluation**	How are success, cost and repeatability checked?
05	**Control**	When does it stop and who is accountable?

The next step is the Chinese ecosystem

Models, platforms and factories connect these architectures to economic decisions.

01

Model

Language and reasoning capabilities.

02

Agent system

Tools, memory and controls.

03

Robot

Perception, control and physical execution.

04

Outcome

Correct tasks, availability and cost.